Research Statement

Kevin Tuttle

My research lies in the field of low-dimensional topology, specifically a study of contact 3 and corresponding symplectic 4 dimensional manifolds they bound. 3- and 4- dimensional manifolds are of interest because they are the lowest dimension where unusual phenomena occur, and 4 dimensions is the highest dimension before algebraic conditions rigorize all structures. Indeed, dimension 4 is the only dimension with uncountably many different smooth structures on $\mathbf{R}^4$. Tools to study these manifolds connect to algebraic geometry, complex analysis, homological algebra, dynamics, and combinatorics.

Contact topology originated as a tool in Lie's work on differential equations, and was revolutionized in the late 20th century/early 21st by the works of Giroux, Eliashberg, and Thurston. Contact topology's sibling, symplectic topology, has its roots in the geometry of a phase space in classical mechanics, and has notable contributions from Weinstein and Fukaya. Walking in their footsteps, my research centers on algorithmic utilization of the Giroux Correspondence, that is, an equivalence between contact 3-manifolds, usually given a topological description via knot surgery diagrams, and open book decompositions of 3 manifolds, which use algebraic notions of mapping class groups.

Bourgeois, Ekholm, and Eliashberg utilized differential graded algebra (DGA) techniques in order to produce invariants of symplectic manifolds with contact manifold boundaries. Such invariants calculate a DGA from a Legendrian surgery presentation of a manifold, that is, a series of framed knots in S^3. Now, Legendrian surgery diagrams represent a contact structure, and thus the Giroux Correspondence transitively equates these surgery presentations with open book decompositions (up to positive stabilization). Thus it was a long standing problem in the community if one can calculate such DGA invariants directly from an open book decomposition. Avdek and Stenhede, among others, worked on the details the conversion process from open book decompositions to surgery diagrams, and my work concerns the algorithmic nature of such.

Specifically, I am interested in the computational process of inputting an open book decomposition into a program, and being able to output a surgery diagram description in "Gompf Normal Form," and from that description further compute invariants of the contact structure. Examples of such invariants include Legendrian contact homology, or the d_3 invariant. Both invariants are algorithmic to calculate, and thus part of my work has been python coding, which can be viewed on my github.

Undergraduate Research. One important tool my research utilizes is knot theory, which in a less hyperspecific setting is a very accessible tool to undergraduates for research projects. Without worrying about the full topological machinery, there exist different homology theories for calculating knot invariants, such as

1. Legendrian contact homology,
2. knot Floer homology (via grid diagrams),
3. Khovanov homology.

In seeking to calculate such invariants students get experience with some topological topics such as knot theory, algebraic concepts such as homology of a DGA or a chain complex, and experience algorithmic processes along the way. Coding experience would be optional depending on the specific direction of the project, but could give students a valuable asset for their resume, and provide real world experience outside the hypotheticals of mathematics. Undergraduates would be able to calculate the invariants for specific families of knots, and thus contribute to the community's knot atlases as part of a thesis.